Varendra University

Department of Computer Science and Engineering

Undergraduate Project Proposal

Developing a Machine Learning-Based System for Real-Time Drowsiness Detection

Submitted by:

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1 Introduction

Drowsy driving is a significant issue that contributes to a large number of road accidents globally. It is a state in which a driver experiences excessive sleepiness or fatigue, leading to reduced alertness and slower reaction times. This dangerous condition poses a serious risk not only to the driver but also to passengers, other road users, and pedestrians. Traditional methods of detecting drowsiness, such as self-assessment or passenger monitoring, are often unreliable and impractical. Therefore, there is a growing need for automated systems that can accurately and efficiently detect driver drowsiness in real time.

This thesis focuses on developing a machine learning-based system for real-time drowsiness detection using live camera feeds. By leveraging advanced techniques in computer vision and deep learning, specifically Convolutional Neural Networks (CNNs), the proposed system aims to monitor and analyze the driver's eye movements to detect signs of drowsiness. The system integrates with a real-time alert mechanism to warn drivers when signs of drowsiness are detected, thereby enhancing road safety. This research not only demonstrates the feasibility and effectiveness of such a system but also highlights the potential of artificial intelligence in preventing road accidents and saving lives.

2 Background Study

Various researchers have contributed significantly to the development of systems for detecting drowsiness in drivers, utilizing a range of methodologies and technologies. The content analysis of selected research work is presented below:

No.	Paper Name	Methodologies	Findings	Limitations
1	Driver Drowsiness Detection System using Convolutional Neural Network	CNNs, Image Processing, Haar Cascades	The CNN-based approach achieved high accuracy in real-time detection, demonstrating its effectiveness in practical applications.	The method may not perform well in low light, with glasses, or when the face is partially covered.
2	A Robust and Efficient EEG-based Drowsiness Detection System	EEG Signal Analysis, Machine Learning Algorithms	The system demonstrated high sensitivity to drowsiness states, but the need for physical sensors limits its practical use.	EEG-based detection requires special sensors, which may not be practical for real driving use.
3	Physiological Signal-based Drowsiness Detection	Multimodal Signal Analysis, Machine Learning	The study reported enhanced detection accuracy and robustness, especially in varying conditions.	The system may need a large dataset and high computational power for better accuracy.

From the above discussion, it is evident that significant advancements have been made in drowsiness detection systems, employing a variety of methods including CNNs, EEG signal analysis, and multimodal data fusion. This work aims to build upon these foundations by developing a robust system that integrates real-time facial analysis with machine learning algorithms, specifically CNNs, to provide accurate and immediate detection of drowsiness. The proposed model will utilize a dataset composed of labeled eye state images and evaluate its effectiveness under various conditions.

3 Problem Statement

Drowsy driving is a major cause of road accidents, but many existing detection methods are unreliable or impractical in real-world driving. Self-assessment and passenger monitoring cannot provide continuous, objective measurement, while sensor-based systems such as EEG often require wearable devices that are uncomfortable and costly for drivers.

Therefore, there is a research gap for a non-intrusive, low-cost, and real-time solution that can accurately detect driver drowsiness using only a camera. This project aims to address that gap by extracting facial and eye features from live video, classifying eye states using CNNs, and triggering an immediate alert to reduce accident risk under different lighting and driver conditions.

4 Objectives

The objectives of our research are given below:

- Develop a machine learning-based system for real-time drowsiness detection using live camera feeds.
- Enhance the accuracy of drowsiness detection by analyzing eye movements and facial features with Convolutional Neural Networks (CNNs).
- Integrate a real-time alert mechanism to promptly warn drivers upon detecting signs of drowsiness.
- Evaluate the system's performance under diverse environmental conditions and with varying facial features to ensure robustness.

5 System Design

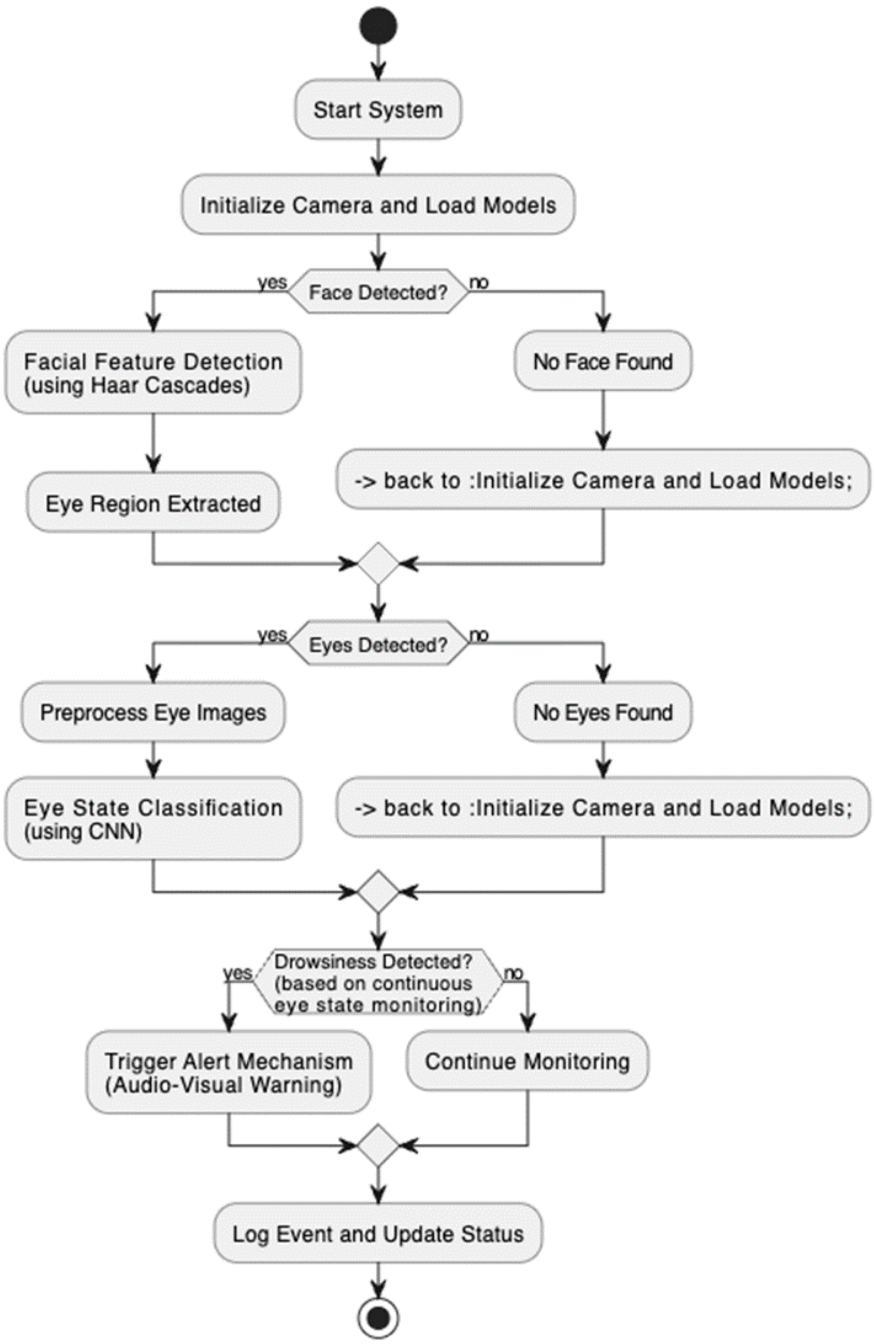

Figure 1: Proposed System Design Drowsiness Detection System

The goal is to develop a real-time drowsiness detection system using machine learning techniques to enhance road safety. As part of this research, we will implement our work by combining the following methods:

1. Facial Feature Detection using Haar cascades to identify and isolate the eye region from live camera feeds.

2. Eye State Classification with Convolutional Neural Networks (CNNs) to distinguish between open and closed eye states.
3. Real-Time Monitoring by continuously analyzing eye movements and facial features to detect signs of drowsiness.
4. Alert Mechanism Integration to provide immediate audio-visual warnings when drowsiness is detected, enhancing driver awareness and safety.

6 Expected Outcomes

Here are expected outcomes tailored to the report:

1. A fully functional real-time drowsiness detection system using live camera feeds that can run on standard computing hardware.
2. A trained CNN model capable of accurately classifying eye states (open/closed) and detecting drowsiness with reported metrics such as accuracy, precision, recall, and F1-score.
3. Experimental results and performance evaluation under diverse lighting conditions and with drivers having different facial features, including confusion matrices and comparative analysis.
4. An integrated audio-visual alert mechanism that triggers timely warnings when drowsiness is detected and demonstrates its effectiveness in simulated or real driving scenarios.

7 Tools and Technologies to Be Used

Programming language: Python (for model training + real-time pipeline).
Libraries: OpenCV (Haar cascades, video capture), TensorFlow/Keras or PyTorch (CNN training/inference), NumPy/Pandas (data handling).
Development tools: Google Colab (experiments), PyCharm (implementation).
Hardware: Webcam, laptop.
Dataset: Labeled eye-state images (open/closed) such as CEW (Closed Eyes in the Wild).

8 References

[1] "Driver Drowsiness Detection System using Convolutional Neural Network," *IEEE Conference Publication, IEEE Xplore*. [Online]. Available: `https://ieeexplore.ieee.org/document/9777182`

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9 Work Division and Team Management

Team Member	**Role**	**Main Responsibilities**	**Key Deliverables**
Md Karim	Project Lead /Coordinator	Planning, timeline tracking, task coordination, final integration oversight	Work plan, progress tracking, final compiled project
Samia Akter Nipa	Data & Preprocessing	Dataset collection/selection, labeling, cleaning, augmentation, train/val/test split	Prepared dataset, preprocessing pipeline
Md Karim	ML Engineer (CNN)	Model design, training, hyperparameter tuning, evaluation metrics	Trained CNN model, results (accuracy, F1, confusion matrix)
Mst. Khadiza Alam Momi, Samia Akter Nipa	Real-time CV & Integration	Haar cascade face/eye detection, camera pipeline, real-time inference integration	Real-time detection module, optimized runtime performance

Mst Khadiza Alam Momi	UI/Alerts & Documentation	Audio-visual alert integration, demo/prototype, report writing and formatting	Alert system, demo, documentation/report

10 Timeline

Time (Week)	Activities
Week 1–4	Planning (1 month)
Week 5–7	Background Study (20 days)
Week 8–9	Data collection & preprocessing (15 days)
Week 10–13	Model development & training (1 month)
Week 14–16	System integration & testing (20 days)
Week 17–18	Performance evaluation (10 days)
Week 19–20	Final report writing (10 days)

11 Cost-Benefit Analysis

Below is a short cost-benefit analysis:

- Software (Python, OpenCV, TensorFlow/PyTorch): 0 BDT (free/open-source).
- Dataset (public eye-state datasets): 0 BDT (using open datasets).
- Hardware: Webcam 1,000–3,000 BDT.
- Compute options (training): Google Colab (Free): 0 BDT (best cost-effective for student research).

12 Constraints and Limitations

Economic: Limited budget for GPU training, paid cloud compute, and high-quality camera hardware.
Environmental/Operational: Performance can drop under poor lighting, night driving, shadows, camera vibration, and different camera angles.
Technical: Haar cascade may fail with partial face visibility, occlusions (glasses/masks), fast head movement; CNN accuracy depends on dataset size/quality and generalization.
Real-time constraints: Need low latency and stable FPS; limited CPU/GPU resources can reduce real-time performance.

Ethical/Privacy: Capturing facial video raises privacy concerns; requires informed consent, secure storage, and avoiding misuse of recorded data.
Health & Safety: False positives may annoy or distract drivers; false negatives are risky - alert design must minimize distraction while remaining effective.

13 Societal Impact

Improves road safety by detecting early signs of driver drowsiness and providing real-time alerts, helping reduce fatigue-related accidents and injuries.

Enhances quality of life for drivers and passengers by supporting safer long-distance travel, especially for night driving and professional drivers (bus/truck/ride-sharing).

Provides a low-cost, non-intrusive alternative to sensor-based systems (e.g., EEG wearables), making safety technology more accessible and scalable.

Supports ethical AI use by encouraging privacy-aware design (consent, minimal data storage, secure handling) while delivering real-world community benefit.

14 Complex Engineering Problem Mapping

Table 3: Mapping with complex problem solving.

WP1	**WP2**	**WP3**	**WP4**	**WP5**	**WP6**	**WP7**
Dept. of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent of Stakeholder Involvement	Interdependence
✓	✓	✓				

15 Complex Engineering Activity Mapping

Table 4: Mapping with complex engineering activities.

EA1	**EA2**	**EA3**	**EA4**	**EA5**
Range of resources	Level of Interaction	Innovation	Consequences for society and environment	Familiarity
✓	✓			

16 Green Computing Considerations

I will use lightweight computer-vision methods (Haar cascade for eye/face detection) and a small CNN model so the system can run efficiently on a normal laptop without heavy power use. I will reduce unnecessary computation by lowering the camera frame rate and resolution when possible and processing only the eye region instead of the full video frame. For training, I will prefer free/shared resources (e.g., Google Colab) and limit training epochs using early stopping to avoid wasting GPU time and energy. I will store only necessary data (or avoid saving video completely) to reduce storage needs and digital carbon footprint.

17 Lifelong Learning and Personal Development

As a student, this project will strengthen my skills in computer vision and deep learning by building a real-time system using Haar cascades and CNNs. I will improve my research ability through literature review, dataset preparation, experimental evaluation, and writing a structured thesis report. I will gain practical experience in system integration (camera input, real-time inference, alert mechanism) and performance optimization. It will also develop my teamwork, communication, and project management skills through task division and regular progress tracking.